

Installing Docker

Introduction

Docker is a platform that allows you to develop, ship, and run applications inside containers. This guide provides steps to install Docker on Windows, macOS, and Linux, using instructions from the official Docker website. We will be going through installing and running the CRISPResso data analysis tool online and locally using Docker. Thus, a working installation of Docker is a prerequisite for running Docker locally. Docker is usually straightforward to install provided your computer has the required privileges – you may need to contact the IT services of your organisation if your computer is locked down to prevent installation of some software, otherwise, installation and testing of a successful Docker install is straightforward. Instructions for installing on each machine are given below.

Common Pre-requisites

- Stable internet connection
- Administrative privileges on your computer
- Basic familiarity with terminal or command-line interface (CLI)

Installing Docker on Windows

1. Visit the official Docker installation page for Windows:
<https://docs.docker.com/desktop/install/windows-install/>
2. Ensure you are using Windows 10 or 11 Pro, Enterprise, or Education (WSL2 is required).
3. Enable virtualization in BIOS and install WSL2. Docker provides a helpful guide here:
<https://docs.docker.com/desktop/windows/wsl/>
4. Download and install Docker Desktop.
5. Restart your computer if prompted.
6. Open PowerShell and type ``docker run hello-world`` to verify installation.

Installing Docker on macOS

1. Visit the official Docker installation page for Mac:
<https://docs.docker.com/desktop/install/mac-install/>
2. Choose the correct version for your system (Apple chip or Intel chip).
3. Download and drag Docker Desktop into the Applications folder.
4. Open Docker Desktop from Launchpad.
5. Open Terminal and run ``docker run hello-world`` to verify installation.

Installing Docker on Linux (Ubuntu/Debian)

1. Follow the Linux installation guide: <https://docs.docker.com/engine/install/ubuntu/>
2. Use terminal to add Docker's official GPG key and set up the stable repository.
3. Install Docker Engine using apt.
4. (Optional) Add your user to the `docker` group to avoid using `sudo`.
5. Run `docker run hello-world` to verify installation.

Final Notes

For complete documentation, troubleshooting, and advanced options, visit the Docker Docs hub: <https://docs.docker.com/>

Additional Resources

Docker Tutorial To Get Started:

<https://www.youtube.com/watch?v=DQdB7wFEygo>

Docker Tutorial for Beginners:

<https://www.youtube.com/watch?v=b0HMimUb4f0>

How to install Docker on Windows - 2025 [step by step guide]:

<https://www.youtube.com/watch?v=ZyBBv1JmnWQ>

How to install Docker on Windows 11 (Updated 2025):

<https://www.youtube.com/watch?v=JBEUKrjbWqg>

Install Docker on Ubuntu:

<https://www.youtube.com/watch?v=cqbh-RneBlk>

Install Docker on MacOS:

<https://www.youtube.com/watch?v=-EXIfSsP49A>

Docker Tutorial for beginners: <https://docker-curriculum.com/>

Install Docker on Windows

<https://docs.docker.com/desktop/setup/install/windows-install/>